VICTORY 2027

FORCE, WORK, POWER & ENERGY

LECTURE-3 PHYSICS

Shailendra sir



Topics To Be Covered

1. UNIFORM CIRCULAR MOTION
2. CENTRIPETAL FORCE
3. WORK
4. QUESTIONS


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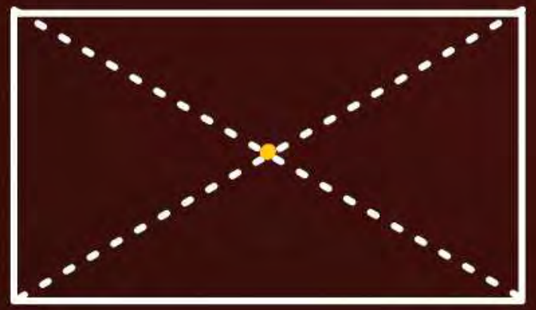

Discipline • Dedication • Success
SSS ARMY
Where Average
Turns Into Topper
Physics Wallah
Discipline • Dedication • Success
Join the Movement

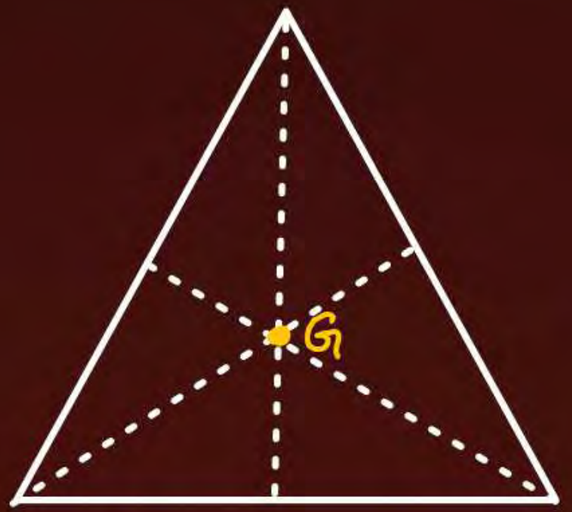
Center of gravity

↳ Weight

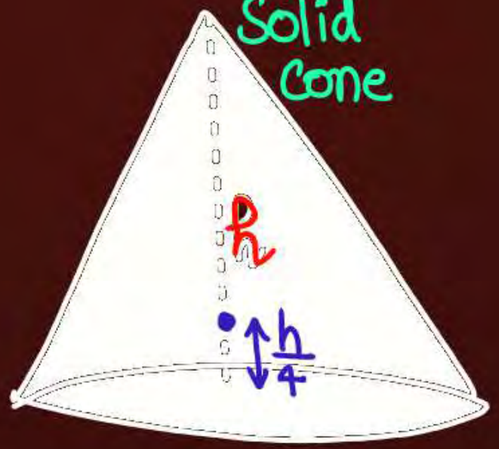




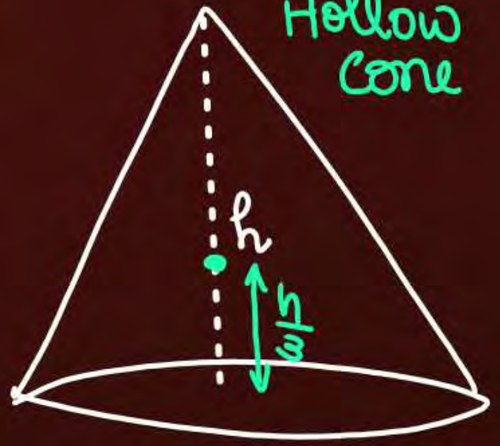




Solid Cone



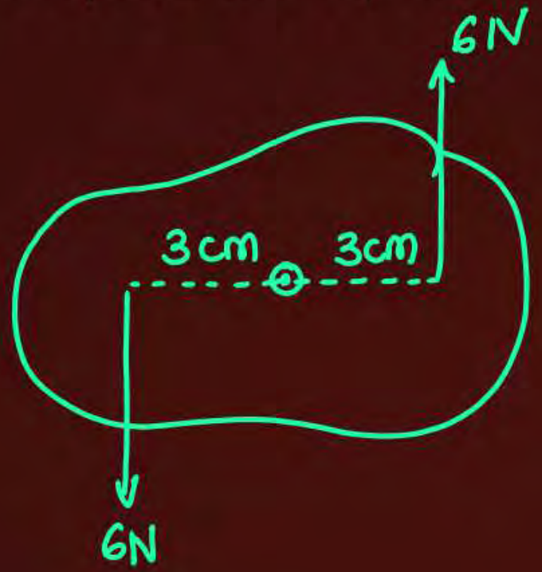
Hollow Cone



QUESTIONS

The resultant moment of couple, if two equal forces each of 6 N are acting at a distance of 3 cm each from the centre is:

1. 36 Nm
2. 3.6 Nm
3. 360 Nm
4. 0.36 Nm



$\tau = r \times F$

$= \frac{6}{100} \times 6$

$= \frac{36}{100}$

$= 0.36$ Nm

3 cm → $\frac{3}{100}$ m

1 m = 100 cm

1 cm = $\frac{1}{100}$ m

UNIFORM CIRCULAR MOTION

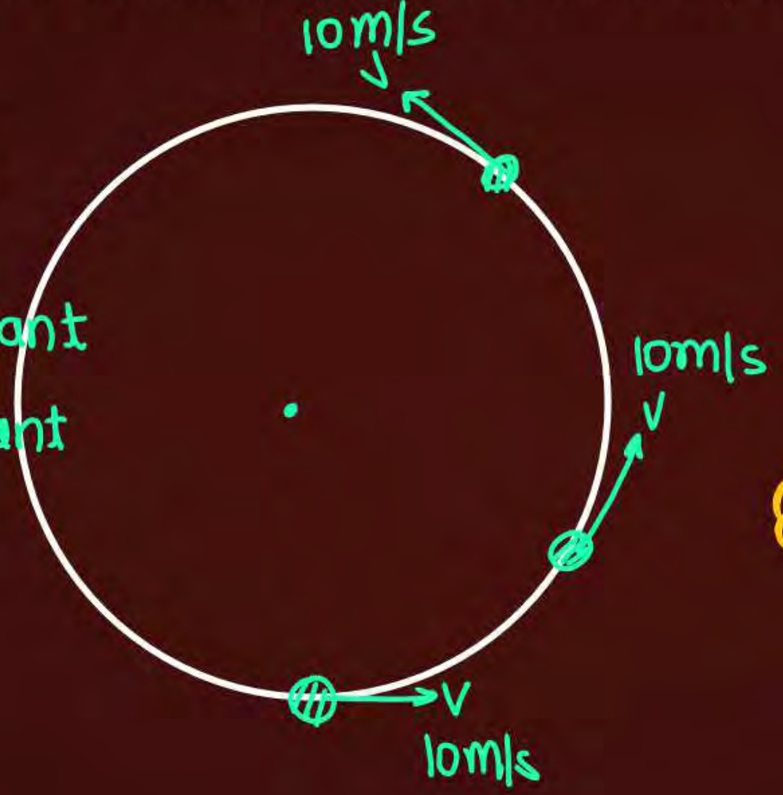
When a particle moves with a constant speed in a circular path, its motion is said to be the uniform circular motion.

V = constant

↳ Speed

Direction ≠ constant

Velocity ≠ constant



Accelerated Motion

$a = \frac{\Delta V}{\Delta t}$

↳ Rate of change of velocity

SSS

Circular Motion is always an accelerated motion.

Non-Uniform Motion



Speed ≠ constant

Direction ≠ "

Velocity ≠ "

SSS

UCM and NUCM both are accelerated Motion.



Centripetal Acceleration → it changes the direction in circular motion.

$$a_c = \frac{v^2}{r}$$

Vector Quantity

m/s^2 or cm/s^2

↓ SI ↓ CGS.



$$a_c = \frac{v^2}{r} = \frac{(10)^2}{5} = 20\,m/s^2$$

Acceleration

↓

it always changes velocity of object.

- Speed
- Direction

UNIFORM CIRCULAR MOTION

always accelerated motion

Unaccelerated Motion
a=0

Difference between the uniform circular motion and uniform linear motion:

speed = const.
Direction ≠ "
Velocity ≠ "

speed = constant
Direction = constant
} Vel. → const.



In **uniform linear motion**, the speed and velocity are constant and acceleration is zero i.e., the uniform linear motion is an **unaccelerated motion**, while in a **uniform circular motion** the velocity is variable (although speed is uniform), so it is an **accelerated motion**.

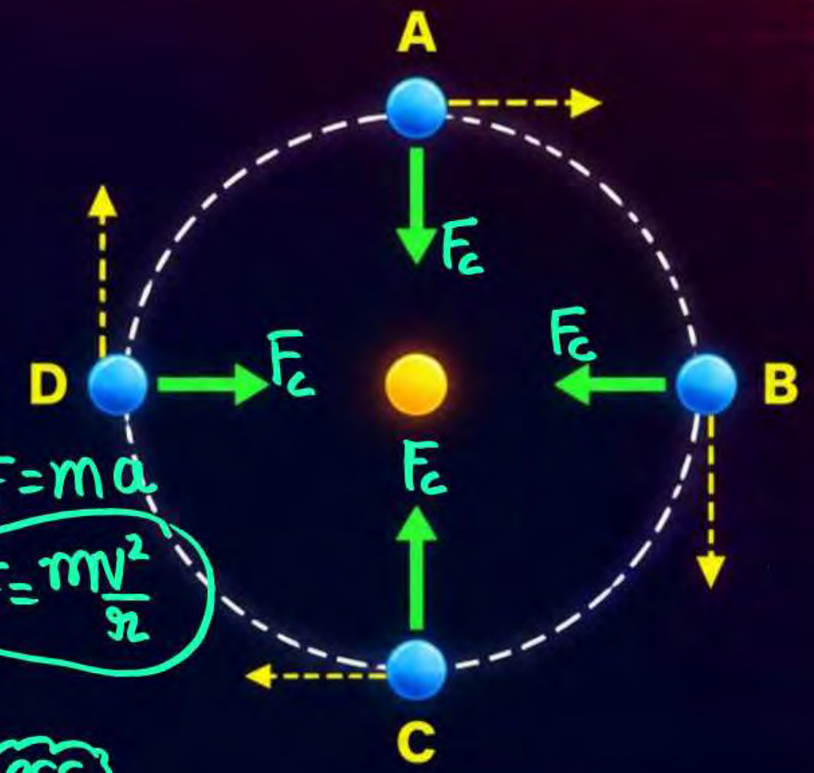
CENTRIPETAL FORCE

In circular motion, at each point of circular path, the particle continuously **changes its direction** of motion.

This change in direction of motion can not be brought without a force. Thus, the motion in circular path is possible only under the influence of a force which is termed as the **centripetal force**.

At each point of circular path, this force is directed **towards the centre** of the circle. Thus the direction of acceleration also changes at each point of the circular path, but its magnitude remains the same i.e., the **acceleration is variable (or non uniform).**

Hence for a body moving in a circular path, a force is needed which acts as the **centripetal force**.



SSS

→ **Centripetal Force:** Acts towards the centre and keeps the body in circular path.

⇢ **Direction of Motion:** Tangential to the circular path.

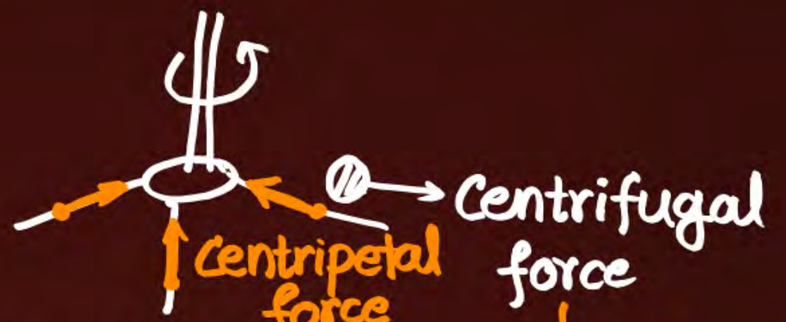


$$a = \frac{\Delta V}{\Delta t} = \frac{V-u}{t}$$

$$F = ma$$

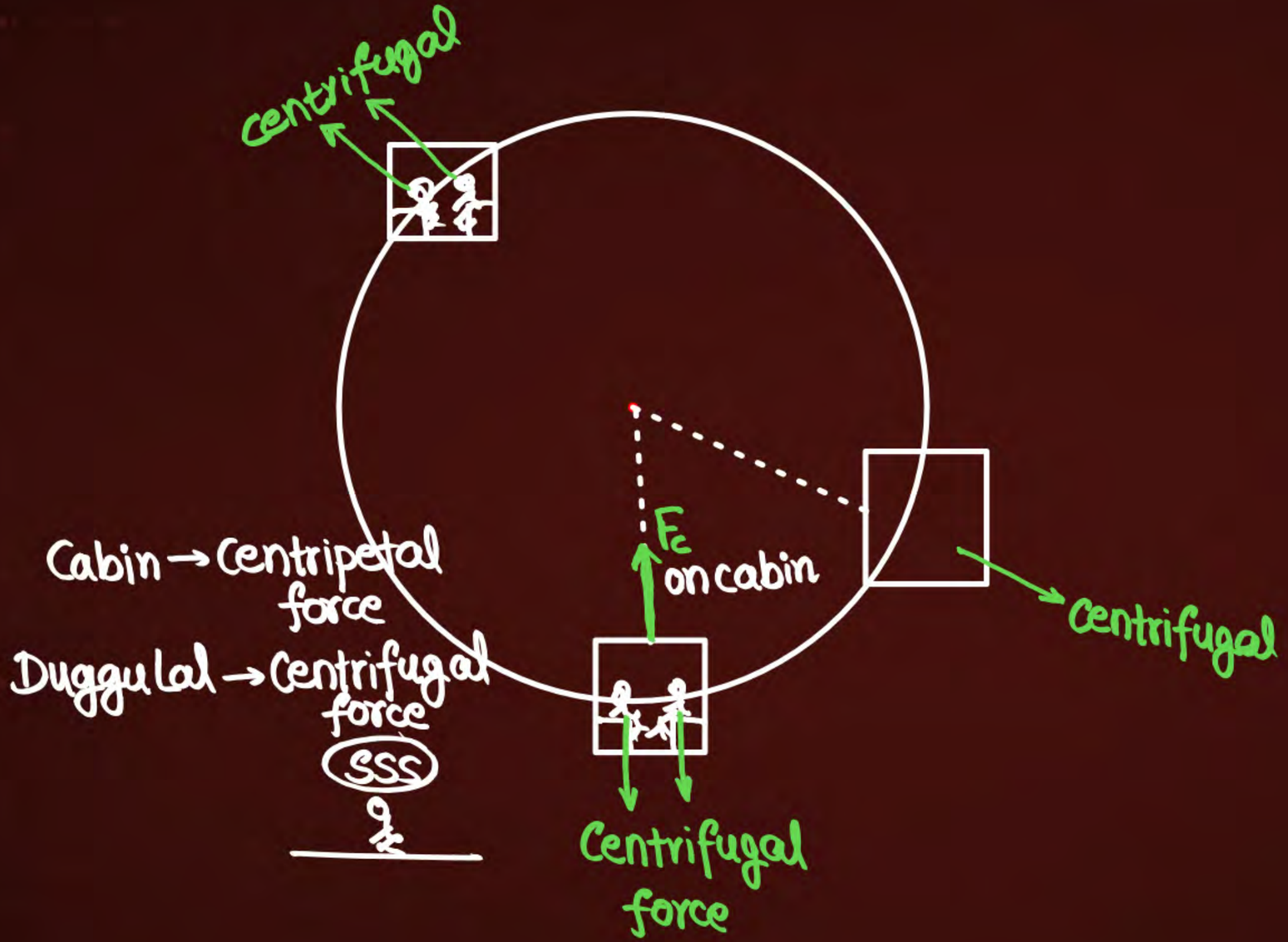
$$F = m\left(\frac{V-u}{t}\right)$$

To change velocity of object.



@ inwards

~~@~~ outwards


centrifugal
Cabin → Centripetal force
Duggu Lal → Centrifugal force
SSS
F_c on cabin
Centrifugal
Centrifugal force

CENTRIFUGAL FORCE

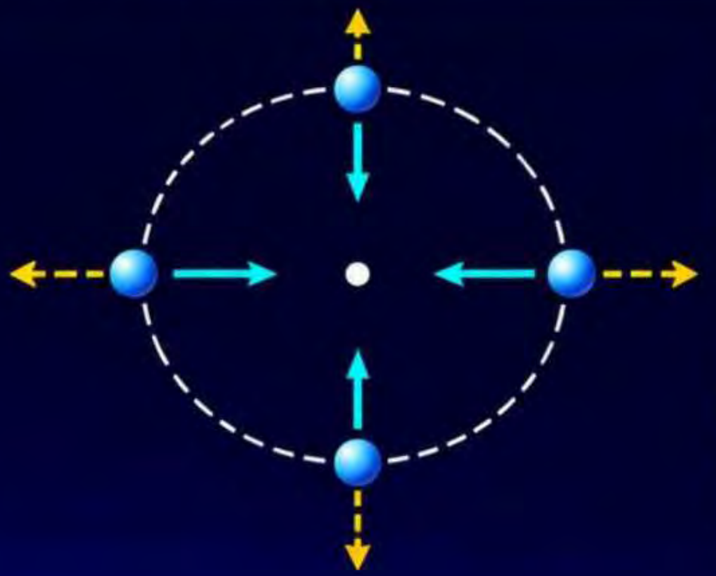
A force acting on a body away from the centre of circular path is called the **centrifugal force**. Thus centrifugal force is in a direction **opposite** to the direction of centripetal force.

Its magnitude is same as that of the centripetal force. But **centrifugal force** is not the **force of reaction** of the centripetal force because action and reaction do **not act on the same body**.

It is not the real force, but is a **fictitious force** assumed by an **observer moving in circular path** alongwith the body.

Centripetal Force (Towards Centre)

Centrifugal Force (Away from Centre)



Centripetal Force: Acts towards the centre and keeps the body in circular path.

Centrifugal Force: Acts away from the centre and opposes the centripetal force.

QUESTIONS

Which of the following quantity remains constant in a uniform circular motion:

1. Velocity
2. Speed
3. Acceleration
4. Both velocity and speed

QUESTIONS

The centrifugal force is:

1. A real force
2. The force of reaction of centripetal force
3. A fictitious force
4. Directed towards the centre of circular path

Work

Chap-2

Work is said to be done only when the force applied on a body makes the body move (i.e., there is a displacement of the body).

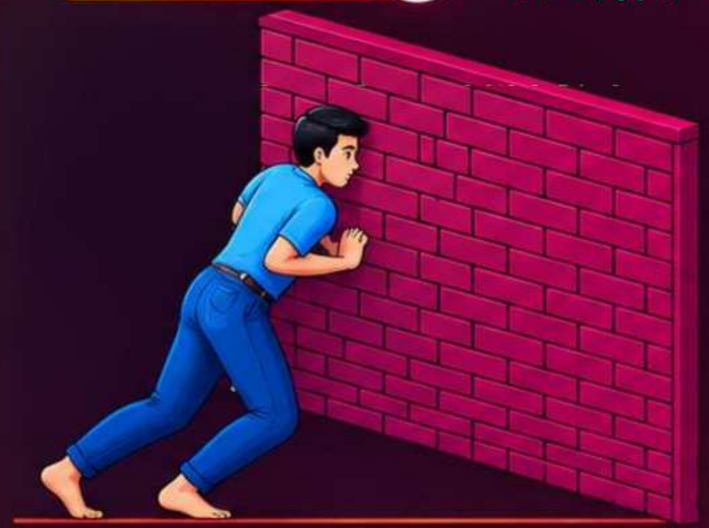
MOTION ✓

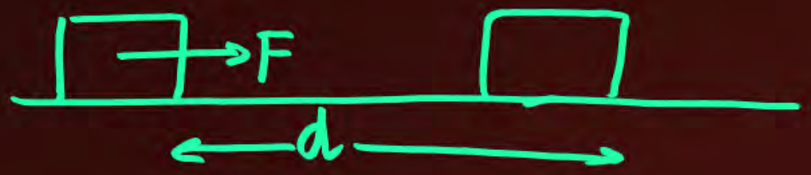
WORK ✓



NO MOTION ✗

WORK ✗



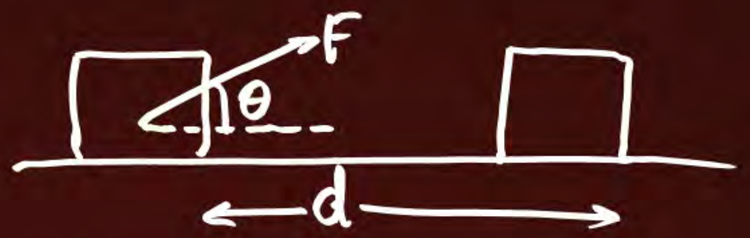


$$\text{Work} = F \cdot d$$

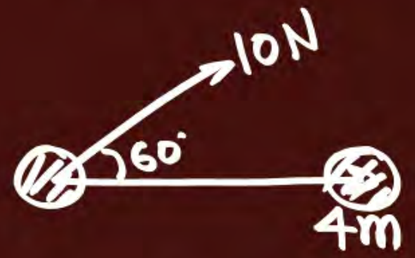
↳ When F and d both are in same direction.



$$W = Fd = 10 \times 4 = 40\ J$$



$$\text{Work} = Fd\cos\theta$$



$$W = Fd\cos\theta$$
$$= 10 \times 4 \times \cos 60^\circ$$
$$= 20\ J$$

$\Rightarrow$

$\cos 0^\circ = 1$

$\cos 30^\circ = \frac{\sqrt{3}}{2}$

$\cos 45^\circ = \frac{1}{\sqrt{2}}$

$\cos 60^\circ = \frac{1}{2}$

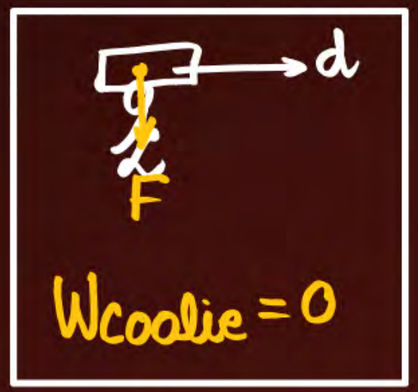
$\cos 90^\circ = 0$

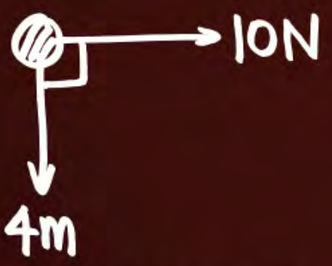
$\cos 180^\circ = -1$

F ✓

d ✓

$\theta \neq 90^\circ$ ✓

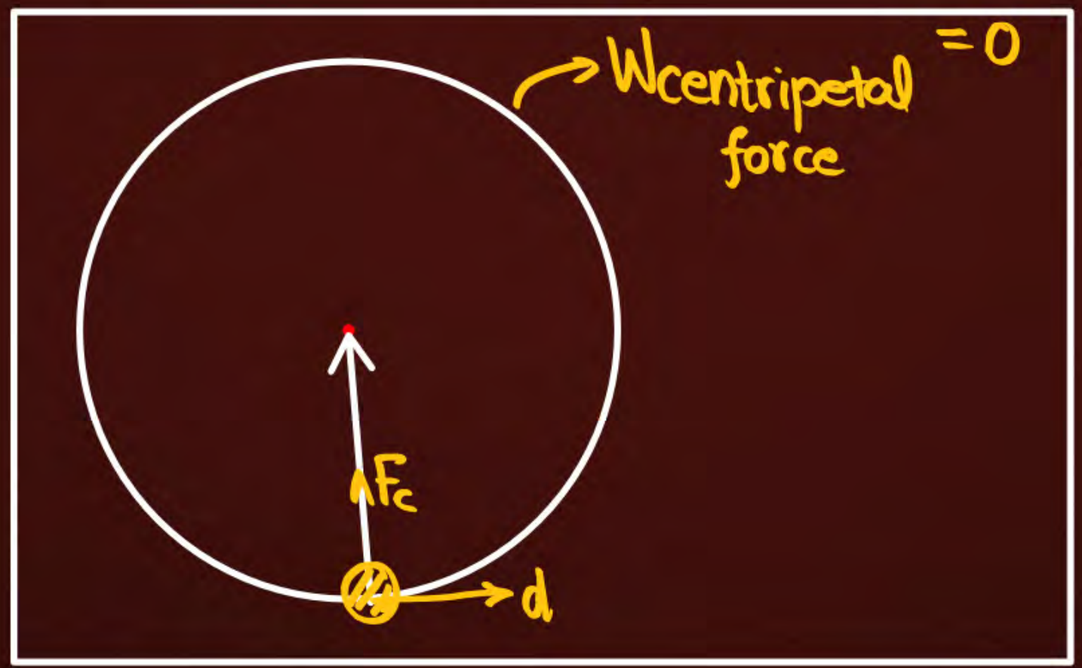




$W = Fd\cos\theta$

$= 10 \times 4 \cos 90^\circ$

$\boxed{W = 0}$



Work

The amount of work done by a force is equal to the product of the force and the displacement of the point of application of the force in the direction of force.



$$W = Fd$$

Scalar Quantity

SI → Nm or Joule

CGS → dyne cm or erg

QUESTIONS

A force of 10 N displaces a body by a distance of 2 m at an angle 60° to its own direction. Find the amount of work done.

$W = Fd\cos\theta$

$= 10 \times 2 \times \cos 60^\circ$

$= 10 \times 2 \times \frac{1}{2}$

$= 10\ J.$

QUESTIONS

In situations wherein the displacement is in the same direction as the force. When this occurs, work is said to be?

1. Zero
2. Positive
3. Negative
4. Finite



$\theta = 0^\circ$

$W = Fd\cos 0^\circ$

$\boxed{W = Fd}$

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You